



TYPES OF **VISUALIZATIONS** IN **POWER BI**

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BAR CHART



Chart that uses rectangular bars to represent data, with the size of each bar proportional to the data value.

Used to represent data that changes over time, to display categorical information.

Eg:

Data of the population changes every year, marks obtained, monthly salary etc.

COLUMN CHART



Chart that displays data in the form of vertical bars that are grouped by category.

Used to compare values across categories or time periods

Eg:

They can be used to show data like sales figures, market share, and customer demographics.

STACKED BAR CHART



Bar charts are means to display a measure against a single dimension.

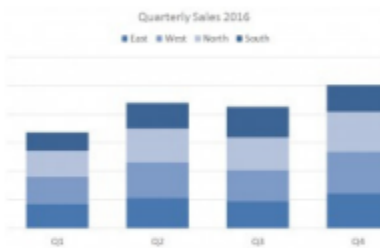
Used to compare multiple categories, show changes over time, or track trends.

Eg:

*Compare the sales of different products in a store.

*Track the number of website visitors over time.

STACKED COLUMN CHART

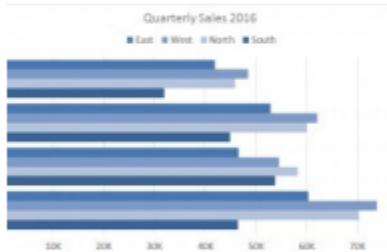


A column chart is a vertical chart, which could present a single bar with multiple stacks in it. used to show the total of two or more categories and to compare the totals and the differences in their components

Eg:

Number of people in the world with and without electricity access from 1998 to 2019.

CLUSTERED BAR CHART



A clustered bar chart, also known as a grouped bar chart or multi-series bar chart.

Used to represent discrete values for more than one item that share the same category.

Eg:

Sales revenue of various departments of the company over several years

CLUSTERED COLUMN CHART

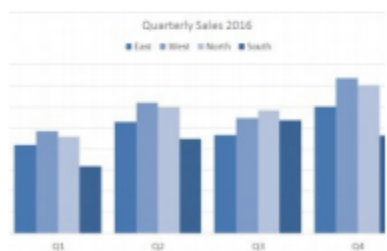


Chart shows multiple data series using grouped vertical bars.

Used to represent discrete values for more than one item that share the same category

Eg:

*Track monthly sales against profit margins.

*Showcase monthly website visitors versus conversion rate.

100% STACKED BAR CHART



Similar to stacked bar charts, but show each series as a proportion of the whole instead of numerical values.

Each stack represents 100%, so you can use color or patterns to distinguish each data series

Eg:

Product market share changes per year.

100% STACKED COLUMN CHART



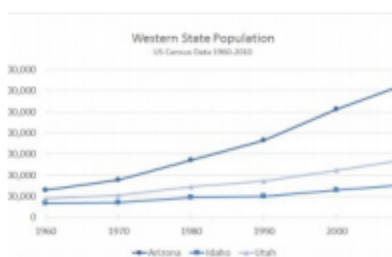
Chart displays columns as wholes with segments indicating the proportions of subcategories within each whole.

Useful for comparing proportional contributions across all categories.

Eg:

A company might use this chart to show which product lines contributed to its revenue by quarter.

LINE CHART



A visual that displays data points as dots connected by lines, and is used to show how data changes over time.

Used to represent large amounts of data over a continuous period of time.

Eg:

Analysis and preparation of government budget, used in medicine and pharmacy to work out the correct strength of drugs.

AREA CHART



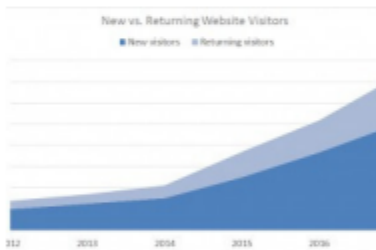
An area chart is a visual representation of data that utilizes both lines and filled areas to convey information.

Used to show how values change over time or across categories.

Eg:

Whether it's sales figures, website traffic, or stock price.

STACKED AREA CHART



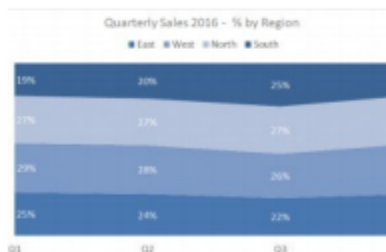
A type of graph that displays the value of multiple groups over time, with each group's value plotted on top of the previous group.

To show how a total value breaks down by groups, and how those groups change over time.

Eg:

Market share of different companies in an industry over a period of time, Carbon emissions over time.

100% STACKED AREA CHART



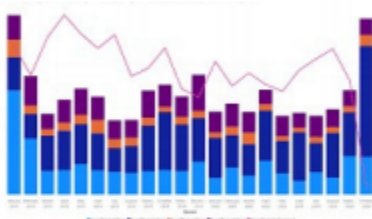
Rectangular chart that shows how the parts of a whole change over time by plotting percentages on the y-axis.

Used to compare the relative contribution of each category to a total over time.

Eg:

A chart can show how the proportion of the world's population that was not in extreme poverty changed from 1820 to 2015.

LINE & STACKED COLUMN CHART



A combination of a line chart and a stacked column chart. It allows you to compare two sets of measures quickly, and can have one or two Y axes.

Used to compare two sets of measures.

Eg: Compare a website's monthly total traffic against its monthly bounce rate. Stacked columns can show the contribution of each traffic source, while the line can show the bounce rate.

LINE & CLUSTERED COLUMN CHART



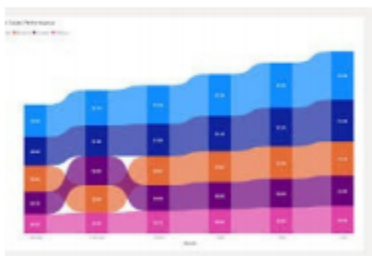
A visualization that combines a line chart and a clustered column chart into one chart.

Used to compare two sets of measures.

Eg:

They can showcase monthly sales of each product (clustered columns) against the overall monthly growth rate (line).

RIBBON CHART



A visual that shows how categories rank over time, and how they contribute to a total. It's also known as a stacked area chart or stream graph.

Used to compare and display data series, trends, and patterns

Eg:

To compare the sales of various products over several months.

WATERFALL CHART



A visual that shows how a series of positive and negative values affect an initial value.

They can be used to represent changes over time, across categories, or for a series.

Eg:

Waterfall charts can be used to plot a company's annual profit, showing the various sources of revenue and arriving at the total profit or loss.

FUNNEL CHART



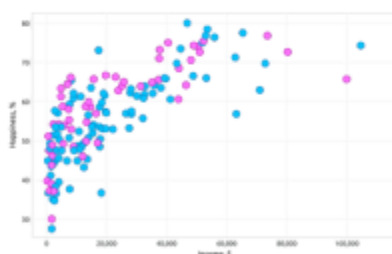
A funnel chart is a graphical representation that shows how data moves through a sequential process.

It's a powerful tool for analyzing and visualizing process flow, especially in sales and project management.

Eg:

Funnel charts can be used to calculate and track conversion and retention rates.

SCATTER PLOT



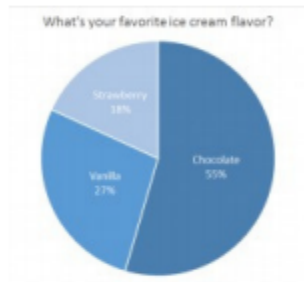
A scatter plot is a chart that uses dots to show the relationship between two variables in a set of data.

Used to observe and show relationships between two numeric variables.

Eg:

You can use a scatter plot to see if there is a connection between ice cream sales and the temperature outside.

PIE CHART

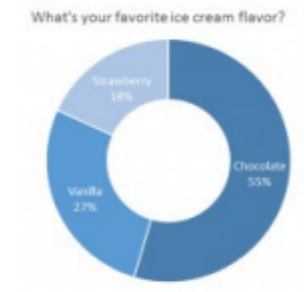


A pie chart is a circular graph that displays data as slices, or wedges, of a pie, where each slice represents a proportion of the whole.

Used to show how a total amount is divided into categories, and are best for comparing the contribution of each group to the whole

Eg: Percentages of the race/ethnicity of public school students across the U.S. for a particular school year.

DONUT CHART



A donut chart is a type of pie chart that features a hole in the center, similar to a donut.

Used to visually represent the relationship between parts and a whole, and to show the proportions of categorical data.

Eg: Donut charts can be used to visualize the causes of lost production time, such as equipment failure, maintenance, shortages, and human error.

TREEMAP



A treemap chart is a data visualization technique that displays hierarchical data in a tree-like structure using nested rectangles.

Useful tool for displaying large amounts of data in a compact and easy-to-read format.

Eg:

Network administrators often use treemaps to analyze disk space use.